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AI-Powered Meal Recognition Recipe Assistant

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Table of Contents

List of Tables

List of Figures

Chapter 1

Introduction & Motivation

Contents

1.1 Introduction

Food is a basic necessity in our lives, but meal preparation is at times challenging for the majority of people. Preparation of meals at home is challenging for many as a result of busy schedules, lack of cooking skills, or lack of adequate meal-planning experience. Much food is wasted due to the fact that people do not have ideas on how to use food they already have. Artificial intelligence (AI) is making cooking easier because technology has transformed everything in life. With AI-driven food recognition, users are able to identify ingredients and find recipes that use their ingredients. This project recognizes ingredients of food and suggests a meal based on that.

Visual and video-capable meal monitoring is increasingly a standard feature of weight management and food consumption monitoring applications. As much as it benefits dietitians and nutritionists by providing useful tools for quick and efficient dietary assessment, it also streamlines the user experience. Rather than managing clunky food diaries or spreadsheet, users are able to conveniently upload a photograph or simply snap short videos of their quinoa salad. The app then comes up with an accurate list—at least an approximation—of its nutritional information, courtesy of a combination of human analysts and machine learning software which improves the accuracy with use..

Image recognition dieting apps provide diverse uses and purposes, having been developed to cater to the needs and objectives of users. A very common use is calorie estimation and nutritional monitoring. Such apps allow individuals to estimate calorific value and nutritional quality of what they have consumed by taking a picture of food. The apps then give a breakdown of macronutrients, such as carbs, proteins, and fats, and micronutrients such as vitamins and minerals, and the percentage of daily recommended intake. Most of these apps also offer users an option to track their intake of foods over time and view how they are doing.

Diet planning and guidance are another significant application. Such applications assist their users in creating their own meal plans and receive guidance and feedback according to their diet objectives, food preferences, and ailments. They can provide them with improved substitutes, recipes, and techniques to enhance the nutritional intake of the user. Furthermore, there are some applications that employ aspects of gamification, reward, and social media to encourage and engage users in their health process..

Lastly, food identification and learning applications enable users to discover about the food they consume, including facts about ingredients, origin, history, and cultural significance. Food applications can also provide details about the environmental and ethical aspects of food production and consumption, such as carbon footprint, water usage, animal treatment, and fair trade. Some of these apps work to increase consciousness and drive action on issues pertaining to food, such as food waste and food security, as well as their function to enhance well-informed food choices.

Nutrition-focused image recognition applications utilize computer vision and artificial intelligence to analyze and process food images captured by the user's device. The image capture is the first stage in this regard, where the user captures an image of the meal via the app camera or loads an image from the device gallery. The application sharpens the image by cropping, rotating, and positioning it such to improve its quality and clarity.

Once the image is prepared, the next process is image segmentation. Here, in this step, the app identifies and delineates various food items in the picture, background items, plates, and cutlery. Each food item, then, gets assigned a bounding box or a mask, meaning they get physically separated from the other items. After segmentation, the process then goes to image classification, where the app classifies each food product with its respective name and category, i.e., pizza, salad, or fruit. Utilizing a pre-trained deep neural network, or a convolutional neural network (CNN), the app matches the food item's features with a large database of food images and labels and returns the most probable label and classification accuracy confidence score.

The final step is image analysis, in which the app determines the portion size, volume, and weight of each food item, as well as its caloric value and nutritional content. This is achieved through various means of portion size measurement, including depth sensors, geometric models, reference objects, and user input. The application then calculates the caloric density and nutritional value of the food product, multiplying these by the portion size estimate in order to accurately calculate the nutritional value. This comprehensive approach enables users to achieve detailed knowledge about their dietary intake, enhancing their ability to effectively control nutrition.

There are numerous advantages of image-based nutrition apps to users. One of them is that it saves time and effort. The apps eliminate users from typing out food names, measuring foods, and weighing foods, which can be very time- and labor-intensive. Users just need to take a photo of their meal, and they get immediate feedback without having to search for the name of the food, select portion sizes, or scan barcodes. Also, these applications are capable of being used offline, hence allowing users to use them even when offline, making it more convenient and accessible.

Another significant advantage is the precision and dependability that these apps bring. They are more accurate and dependable in their caloric value and nutritional estimates compared to the use of traditional methods, such as food labels, nutrition fact charts, and online calculators. These apps are able to recognize a broad variety of foods, such as mixed dishes, home-prepared foods, and ethnic foods, which are usually hard to calculate and sort out. Besides, they also cater to variations in food preparation, cooking, and serving portion sizes, all of which can impact nutritional quality and quantity and thus help users attain a fuller image of their food intake.

Lastly, these applications get users more aware and educated regarding food that they consume and the impact of their diet on their overall health and wellness. These applications help users identify nutritional deficiencies and imbalance in their diet and the source of calories, macronutrients, and micronutrients. Furthermore, these applications can provide helpful information that allows users to make proper dietary decisions, eventually resulting in healthier eating habits and improved nutrition.

1.2 Motivation

The creation of our AI-Powered Meal Recognition Recipe Assistant is a profound appreciation of several intersections in contemporary nutrition and healthcare. We are motivated by three fundamental pillars: technological opportunity, societal need, and personal health empowerment.

Computer vision and natural language processing is growing at an unprecedented rate which provides a fresh chance at transforming how people interface with food. The power of modern AI can now tackle previously unsolvable issues - ranging from highly accurate visual recognition of intricate meals to differentiating between complex nutritional needs. However, what excites us the most is the ability to refine these technologies by Creating personalized models, which not only identify food, but also contextually interpret its significance on an individual's health.

The science of nutrition is quite complicated and, at times, contradictory, leaving consumers uncertain. Guiding AI to remove the ambiguity slows down with evolution fusion of modern scientific nutrition and actionable personalized guidance. Not only does the system recognize items, but it counsels users on their selections, aiding in the establishment of permanent healthy habits instead of offering only short-term tracking solutions.

Lastly, the challenge of making an actual difference in quality of life is what motivates us to develop technology. Unlike most AI technologies that are developed for business or entertainment, our project addresses one of humanity's most primal issues: hunger. Being able to create something that would deeply affect so many people's lives is incredibly motivating. This reason is a complement to our strong belief that, when used responsibly, technology has the potential to improve eating—one of life's most basic activities—for humanity by making it healthier, better informed, and more enjoyable.

1.3 Problem Statement

With the busy nature of life these days, keeping oneself well-nourished has become more difficult for a range of good reasons. First, busy working hours and household tasks leave one with little time for planning and preparing meals in busy lives. Second, excessive conflicting nutrition information provided creates more confusion than clarity, and therefore people find it difficult to make well-informed food choices. Third, traditional methods of monitoring food consumption—such as manual writing or generic calorie-tracking software—become cumbersome and unsustainable in the long term because they're not tailored to the individual.

Existing digital nutritional resources are lacking by providing generic information that ignores important individual factors. Most apps fail to consider portion sizes, cooking methods, or specific health conditions, while recipe websites rarely account for dietary restrictions or personal health goals. This one-size-fits-all approach leads to user frustration and discontinued use.

We address these challenges with a clever system that combines computer vision and machine learning. Our technology correctly identifies meal ingredients from images. Beyond mere recipe suggestions, our system considers nutritional needs, health profiles, and individual preferences to offer genuinely personalized meal suggestions, along with cooking directions and allergen information.

Chapter 2

Related Work

Contents

2.1 Introduction

This section highlights the most similar applications to our meal recognition application, as many existing systems utilize machine learning and computer vision techniques to identify food items from images, offering functionalities such as meal and ingredients recognition, meal instructions, calorie estimation, and other features. These capabilities provide valuable insights for users aiming to maintain a healthy lifestyle. Reviewing these applications helps us identify effective strategies and limitations, guiding the development of our own application to ensure it is both practical and user-friendly.

2.2 AI Photo Recipe Identifier

The *AI Photo Recipe Identifier* app is designed to identify recipes from photos using advanced AI technology. Images of dishes or ingredients are analyzed, and corresponding recipes are provided. Custom recipes can also be generated based on detected ingredients, making it a valuable tool for culinary exploration.

The first step in using the application involves prompting the user to select an image, either by capturing a new photo or uploading an existing one. Accordingly, the initial features offered are "Take Photo" and "Upload Photo." If the user selects the "Take Photo" option, the application provides an additional "Retake" feature, allowing the user to recapture the image before proceeding. This ensures that a clear and suitable photo is obtained for the analysis process. Once satisfied with the image, the user can proceed by selecting "Use Photo" to continue with recipe identification.

Regardless of the method used to provide the image, the application then analyzes it and presents relevant information, including:

- The name of the meal or a list of identified ingredients, along with a brief description.
- A detailed list of ingredients associated with the meal.
- Step-by-step cooking instructions. If the image contains only ingredients rather than a full dish, the application recommends custom recipes based on the detected ingredients.

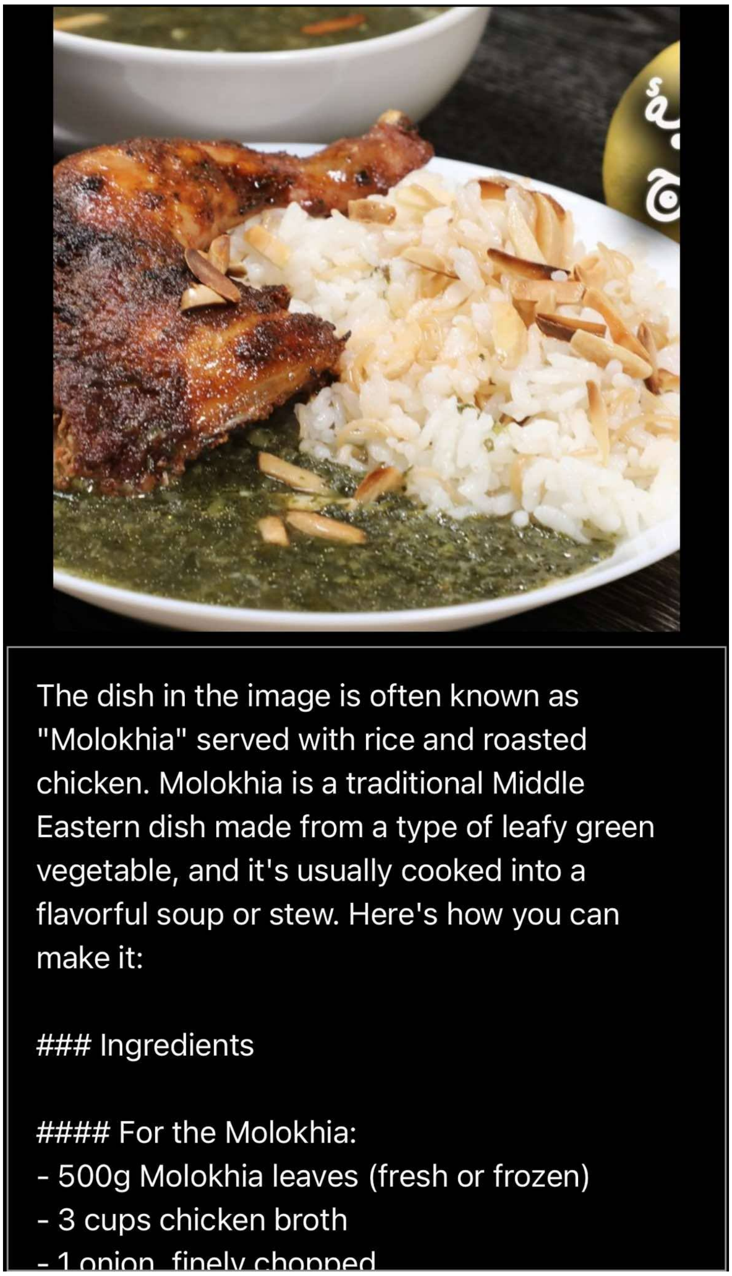

Figure 2.1: An example of AI Photo Recipe Identifier meal recognition.

2.3 ChefMate AI

The *ChefMate AI* application is designed to recognize meal components by scanning an image of the dish. Once the ingredients are identified, the app processes them to suggest a variety of meals that can be prepared using those ingredients. Users are then able to select the most accurate dish from the presented options, after which the app generates a detailed recipe for that specific selection.

ChefMate AI offers several key features that enhance user interaction and functionality:

- **Camera Scan** – captures real-time photos of meals for analysis.
- **Retake Picture** – allows users to retake a photo before processing.
- **Picture Check** – verifies the clarity and usability of the selected image.
- **Generated Recipes** – provides a list of meal suggestions based on detected ingredients.
- **Recipe Description** – displays detailed cooking instructions and ingredient information.
- **Add to Favorites** – enables users to save preferred recipes for quick access.

2.4 ingreGenius

The *ingreGenius* application begins by prompting users to either sign in or create a new account to access its features. Once authenticated, users can scan a meal using their device's camera to identify its ingredients. By selecting the "Identify Meal" option, the app processes the image and generates a comprehensive report containing key nutritional and culinary details. The generated report typically includes the name of the dish, a brief description, estimated calorie count, cooking duration, nutritional information, a complete list of ingredients, and step-by-step recipe instructions. Additionally, users can save their favorite meals within the app for future reference.

The application's key features include:

- **Camera Scan** – captures meal images for ingredient analysis.
- **Generated Recipes** – provides meal suggestions based on identified ingredients.
- **Cooking Time** – estimates the time required to prepare the meal.
- **Calories** – calculates the calorie content of the identified meal.
- **Detailed Ingredients** – lists all components used in the dish.
- **Recipe Description** – includes detailed cooking instructions.
- **Save Recipes** – allows users to store recipes for later use.
- **Add to Favorites** – offers quick access to preferred meals.

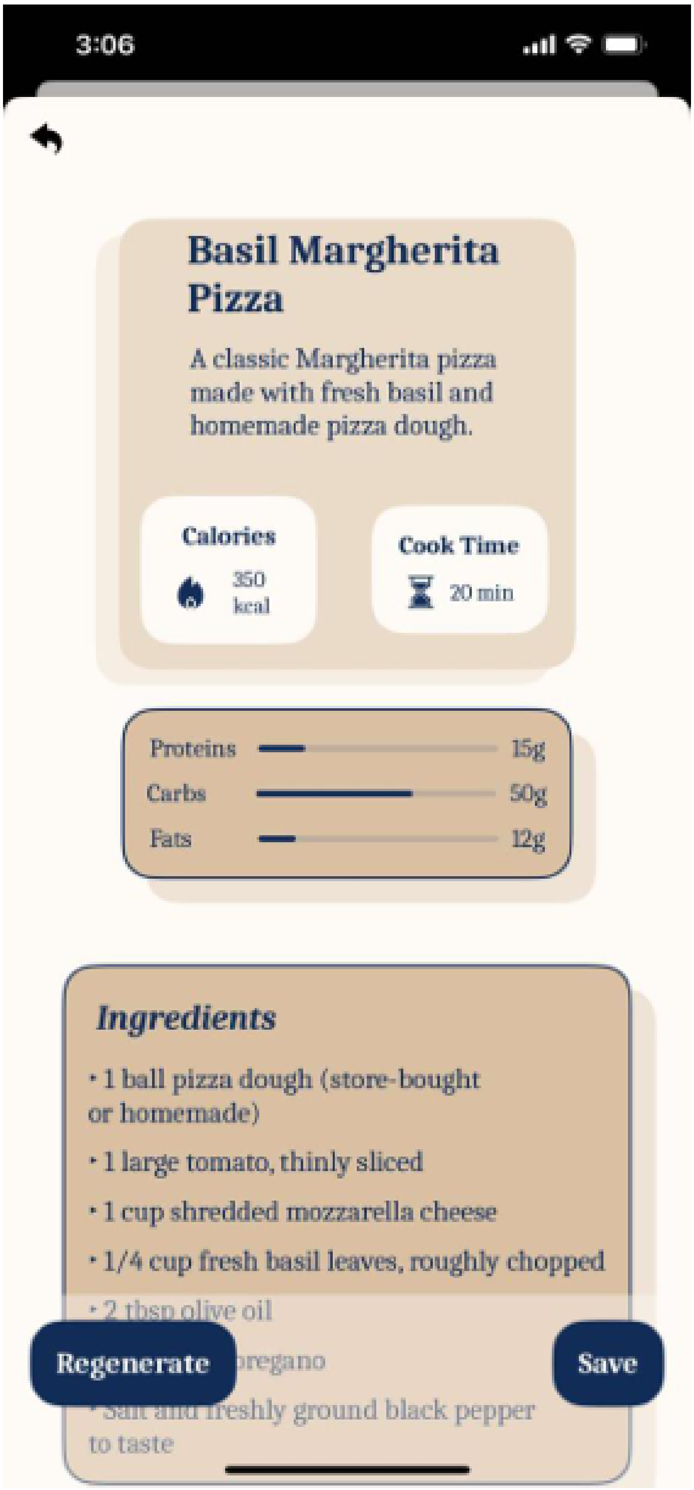

Figure 2.2: Pizza meal recognition by ingreGenius

2.5 SideChef

SideChef is an AI-powered cooking assistant application aimed at making meal preparation easier and more accessible for users of all skill levels. It offers step-by-step recipe instructions, interactive cooking support, and personalized meal planning. By combining user-friendly features with artificial intelligence, SideChef empowers individuals to cook confidently and efficiently, even if they are beginners in the kitchen.

The application's core features include:

- **Step-by-Step Recipes** – clear instructions to guide users through cooking.
- **Ingredient-Based Recipe Search** – allows users to search recipes using available ingredients.
- **Meal Planning** – enables users to organize weekly or daily meals.
- **Voice Command Support** – facilitates hands-free interaction during cooking.
- **Dietary Preferences** – offers filters for specific diets such as vegetarian, keto, or gluten-free.
- **Edit Recipes** – lets users customize recipe steps and ingredients.
- **Cooking Timers** – built-in timers to help manage cooking time effectively.
- **Save Favorite Recipes** – allows bookmarking of preferred meals.
- **Community Features** – lets users share and explore recipes from other home cooks.

User Capabilities:

- Discover new and diverse recipes.
- Cook confidently with guided support.
- Customize and edit recipes based on personal needs.
- Plan meals in advance to stay organized.
- Save time in meal preparation and shopping.
- Share recipes with the cooking community.

Benefits to Users:

- Simplifies the overall cooking process.
- Helps users manage their time more efficiently.
- Contributes to reducing food waste through smart planning.
- Promotes healthy eating habits with personalized options.
- Builds culinary confidence, especially for beginners.

2.6 FOODAI

FOODAI is an AI-powered calorie tracker and nutrition assistant designed to analyze food and deliver comprehensive nutritional insights. By leveraging artificial intelligence, the application assists users in monitoring their calorie intake, understanding the nutritional value of their meals, and making informed dietary decisions. It is particularly useful for individuals who aim to maintain a healthy lifestyle or achieve specific dietary goals.

Key Features:

- **Food Scanning** – detects food items from images for analysis.
- **Calorie Tracking** – monitors daily calorie intake.
- **Nutritional Information** – provides detailed nutritional data per meal.
- **Meal History** – records and stores previously scanned meals.
- **AI Nutritionist** – offers personalized nutritional advice.
- **Dietary Preferences** – allows customization based on user diet types.
- **Recipe Suggestions** – recommends meals based on scanned food items and dietary needs.

User Capabilities:

- Track and manage daily calorie consumption.
- Analyze nutritional content of meals.
- Receive personalized health and dietary advice.
- Explore new and suitable recipe suggestions.
- Maintain a history of meals for progress tracking.

App Functionalities:

- Performs AI-driven food analysis from images.
- Calculates and tracks caloric content.
- Provides real-time nutritional insights.
- Delivers customized dietary recommendations.
- Supports users with intelligent meal planning and goals.

User Benefits:

- Simplifies the process of calorie tracking.

- Encourages healthier eating habits.
- Offers personalized nutrition planning.
- Promotes dietary variety.
- Helps users meet and sustain dietary objectives.

2.7 Summary and Comparative Analysis

To facilitate a clearer comparison and enhance the analysis of the previously discussed applications, Table 2.1 summarizes and contrasts the key features offered by each application. By organizing this information in a structured format, it becomes easier to identify shared functionalities, unique offerings, and potential gaps, ultimately aiding in the evaluation of how these applications relate to and inform the development of our own meal recognition system.

Table 2.1: Comparison of key features across selected meal recognition and nutrition applications.

Feature	**AI Photo**	**ChefMate AI**	**ingreGenius**	**SideChef**	**FOODAI**
Take Photo	✓	✓	✓	✓	✓
Upload Photo	✓	✓	✓	✓	✓
Image Recognition	✓	✓	✓	✓	✓
Ingredient Identification	✓	✓	✓	✓	✓
Recipe Suggestions	✓	✓	✓	✓	✓
Nutrition Information	✗	✗	✓	✓	✓
Step-by-Step Instructions	✗	✗	✓	✓	✓
User Preferences	✗	✓	✓	✓	✓
Shopping List	✗	✗	✗	✓	✗
Voice Assistance	✗	✗	✗	✓	✗
Social Sharing	✗	✗	✗	✓	✗
Photo Cropping/Editing	✓	✗	✗	✓	✗
Photo History	✓	✗	✓	✓	✓
Cooking Time	✗	✗	✓	✓	✓
Edit Recipe	✗	✗	✗	✓	✗
Save Recipes	✓	✓	✓	✓	✓
Allergen Detection	✗	✗	✗	✗	✗
Confidence Score	✗	✗	✗	✗	✗
Food Type Classification	✗	✗	✗	✗	✗
Food Pairing Suggestions	✗	✗	✗	✗	✗

Chapter 3

Background

Contents

3.1 Dataset

3.1.1 Recipe1M+

Recipe1M+ is a large-scale, multidimensional corpus that is specifically designed to take the limits of AI-based meal recognition and recipe assistance to a new level and hence is well-suited for the "AI-powered Meal Recognition Recipe Assistant" project. It contains over one million specially crafted cooking recipes along with approximately 13 million images of food, collected in two phases: initially from popular cooking websites (returning 1 million recipes and 800,000 images) and then with web-sourced images by leveraging Google's image search engine. This huge dataset surmounts the variability of recipes for food, both textually (e.g., ingredients, preparation procedures) and visually (e.g., different presentations of the same dish), and forms a good foundation to train deep models for meal identification from images and related recipe generation. The data is divided into two layers: Layer 1 holds textual data such as recipe names, ingredients, amounts, units, and instructions, with nutritional information for 50,637 recipes from the USDA database; Layer 2 holds the corresponding RGB JPEG images. With 70% of data used for training and the rest split between validation and test sets, Recipe1M+ supports the development of deep learning models for such uses as image-to-recipe retrieval, a crucial task in meal recognition. With diversity and size, the project is able to create an AI companion that can consistently identify dishes through images and provide accurate recipe instructions, enhancing user interaction with cooking content

3.1.2 Arabic Food 101

The "Arabic Food 101" dataset features comprehensive information on 16 renowned and traditional Jordanian dishes, showcasing the extensive richness of Arabic cuisine. It portrays local ingredients, cooking styles, and cultural factors, rendering it a valuable database for food analysis. In project, "AI-Powered Meal Recognition Recipe Assistant," this data set plays a fundamental role in training machine learning models to effectively identify Jordanian meals and suggest original recipes. With the use of structured ingredient and preparation method data from "Arabic Food 101," the AI system can further improve its recognition of dishes based on images and offer context-specific cooking instructions. The public domain license (CC0) of the dataset also allows for easy integration into the project, promoting the objective of combining AI technology with conventional cooking knowledge.

3.2 APIs

In software development, APIs—short for Application Programming Interfaces—play a big role in connecting different systems. They allow one program to ask another for data or services, like when a mobile app gets recipe information from an online source. Instead of building everything from scratch, developers use APIs to save time and keep their code organized. A good example would be using a food recipe API to display meal details without storing the entire database yourself. APIs simplify communication between tools and services, making it easier to build smart, connected applications. Overall, they're a practical way to make modern programs work smoothly together. There are many APIs that are related to food recipes and instructions in the internet such as Spoonacular API, Edamam Food and Recipe API, Yummly API, TheMealDB and many others.

In the following sections, two of these APIs will be discussed in detail.

3.2.1 Spoonacular API

The spoonacular API is a universal food and recipe that allows developers to build applications capable of searching for recipes, retrieving recipe detailed ingredients and cooking instructions, as well as analyzing nutritional information. The spoonacular API has advanced features such as meal planning, diet filtering (e.g., vegan, gluten-free), and food-related data such as wine pairings.

Spoonacular uses a custom-built food database that combines information from:

- Public recipe websites (like *Foodista*, *Taste of Home*, etc.)
- USDA FoodData Central for nutrition info
- Their own curation and user-submitted content

3.2.2 mealDB

TheMealDB is a free and open online recipe database that provides access to a lot of meals from around the world through a simple and developer-friendly API. It offers detailed information for each recipe, including the meal name, ingredients with measurements, preparation instructions, cuisine type, category, images, and even YouTube video links for visual guides.

3.3 Model

3.3.1 Recipe1M+

Dataset Description

This dataset comprises a comprehensive collection of over 1 million recipes, each annotated with detailed metadata. The available data fields include structured ingredient lists with precise measurements (e.g., "2 teaspoons of salt"), step-by-step cooking instructions, high-resolution images of prepared dishes, and supplementary information such as cuisine type, dietary classifications, cooking duration, and estimated difficulty level. The richness of this dataset enables sophisticated computational analysis of culinary content across multiple dimensions.

Model Architecture Overview

The developed system employs a multimodal deep learning approach that synergistically combines natural language processing (NLP) and computer vision techniques. This integrated framework is specifically designed to process and analyze the heterogeneous nature of recipe data, where textual information and visual content provide complementary insights into culinary characteristics.

Natural Language Processing Component

The NLP pipeline begins with advanced text normalization procedures that standardize recipe text into a consistent format. This preprocessing stage handles various lexical variations (converting "tbsp" to "tablespoon"), parses ingredient quantities into structured representations, and tokenizes preparation notes. For semantic understanding, the system utilizes a fine-tuned BERT transformer model that generates contextual embeddings. These embeddings effectively capture nuanced relationships between ingredients, cooking techniques, and culinary styles, enabling tasks such as ingredient substitution analysis and recipe similarity assessment.

Computer Vision Component

The visual processing subsystem employs a ResNet-50 convolutional neural network, pre-trained on ImageNet and subsequently fine-tuned using recipe-specific imagery. This architecture extracts discriminative visual features including dish presentation style, ingredient texture characteristics, and color distribution patterns. The image embeddings complement the textual data by providing visual confirmation of recipe attributes that may be ambiguous in text form alone.

Multimodal Fusion and Knowledge Integration

The model's fusion architecture combines the textual and visual embeddings through a carefully designed neural network layer. This integration enables sophisticated cross-modal applications such as image-to-recipe retrieval and visually-aware recipe recommendation. Additionally, the system incorporates a knowledge graph constructed from recipe metadata that encodes relationships between ingredients, cooking methods, and regional cuisines. This structured knowledge supports complex queries, including dietary adaptation suggestions and culturally-aware recipe modifications.

System Implementation and Optimization

The entire framework is implemented using PyTorch with distributed computing capabilities to handle the scale of the dataset. The inference pipeline has been optimized for low-latency operation, making it suitable for real-time applications in digital cooking assistants and intelligent meal planning systems. The model architecture employs quantization techniques and efficient batch processing to maintain performance while minimizing computational resource requirements.

Applications and Use Cases

The developed system enables numerous practical applications in the culinary domain, including: personalized recipe recommendations based on user preferences and dietary restrictions; automated meal planning with nutritional balancing; intelligent cooking assistants that provide real-time guidance; and novel recipe development through computational analysis of flavor pairing patterns and culinary trends.

3.3.2 Semantic-Consistent and Attention-based Network (SCAN)

This model represents a significant advancement in cross-modal retrieval systems, specifically tailored for aligning food images with their corresponding recipes. At its core, the model tackles two fundamental challenges in food computing: the high visual similarity between dishes from different recipes (large intra-class variance) and the difficulty in extracting meaningful features from noisy, unstructured recipe text. By integrating computer vision and natural language processing techniques, SCAN bridges the modality gap between visual and textual food data, enabling more accurate and semantically meaningful retrieval.

A key innovation of SCAN lies in its sophisticated recipe encoding mechanism. The model processes recipes through two parallel pathways: one for ingredients and another for cooking instructions, both enhanced with self-attention layers. This architecture automatically identifies and emphasizes the most discriminative components of a recipe, such as key ingredients or crucial cooking steps, while downplaying common or irrelevant elements. For instance, when analyzing a strawberry salad recipe, the model might assign higher weights to "strawberries" and "whipped cream" while reducing attention on ubiquitous ingredients like "salt" or "water." This attention mechanism operates without requiring additional supervision or visual cues, making it both efficient and practical for large-scale applications.

The visual processing branch of SCAN utilizes a CNN backbone to extract rich features from food images. What sets the model apart is its novel semantic consistency loss function, which aligns the probability distributions of image and recipe embeddings in the shared feature space. By minimizing the KL divergence between these distributions, the model ensures that semantically similar food items—regardless of modality—are positioned close together in the embedding space. This approach effectively reduces the intra-class variance problem, where visually dissimilar images of the same dish or textually different recipes for similar dishes might otherwise be poorly matched.

Training SCAN involves a multi-task objective combining both the semantic consistency loss and traditional triplet loss. The triplet loss pulls matching image-recipe pairs together while pushing non-matching pairs apart in the embedding space, while the semantic consistency loss provides additional regularization by enforcing probability distribution alignment. This dual-loss strategy leads to more robust embeddings that perform well even when dealing with the inherent noise and variability in real-world food data. The model's efficiency is notable, achieving state-of-the-art performance with just 40 training epochs compared to the 220 epochs required by previous approaches.

When evaluated on the large-scale Recipe1M dataset, SCAN demonstrates superior performance across multiple metrics. It achieves significant improvements in recall rates, particularly for challenging cases where dishes share similar ingredients but differ in preparation methods, or when recipes contain substantial irrelevant text. The model's attention mechanism proves especially valuable in these scenarios, as it can focus on the most distinctive aspects of each recipe. Qualitative results show that SCAN can retrieve appropriate images even when recipes are modified—for example, correctly adjusting results when key ingredients are omitted from the query.

3.3.3 AdaptaFood

AdaptaFood represents a significant advancement in food computing, offering an intelligent solution to adapt recipes based on dietary restrictions such as allergies, medical conditions, or lifestyle choices like veganism. The system is designed to process inputs in two primary formats: recipe images and textual recipe data. For images, it employs a fine-tuned image-captioning model to extract ingredients, while for text, it directly parses and standardizes ingredient lists. By leveraging an attention-based BERT language model, AdaptaFood identifies semantic relationships between ingredients, enabling it to suggest contextually appropriate substitutions that adhere to user-defined constraints. This multimodal approach ensures flexibility and broad applicability, making it a versatile tool for both home cooks and professional nutritionists.

The technical methodology behind AdaptaFood is built on three key pillars. First, multimodal input processing allows the system to handle diverse data sources. For images, a Vision Transformer (ViT)-GPT-2 model generates descriptive captions of ingredients, while textual inputs undergo standardization to ensure consistency. Second, semantic embeddings play a crucial role: a fine-tuned BERT model, retrained on recipe instructions, encodes ingredient semantics to capture nuanced relationships, such as pairing blueberries with dairy products. Sentence-BERT (SBERT) further refines this process by aggregating token-level embeddings into cohesive representations, optimized through max-pooling for enhanced accuracy. Third, knowledge integration involves mapping ingredients to the Composition of Foods Integrated Dataset (CoFID), a comprehensive nutritional database. This enables AdaptaFood to enrich substitutions with detailed dietary information, such as protein or calorie content, and rank them based on cosine similarity and user constraints.

Among its key contributions, AdaptaFood stands out as the first system capable of multimodal recipe adaptation, seamlessly processing both images and text. The retraining of BERT on recipe-specific texts yielded a 15%improvement in substitution quality compared to general-purpose models, highlighting the importance of domain-specific fine-tuning. Practical applications of the system include generating viable substitutions, such as tofu for chicken in vegan diets, which achieved a user satisfaction score of 4 out of 5 in evaluations. These results underscore the system's potential to support dietary planning and promote healthier eating habits.